						1	2	3		4	5	1	2			3	1	2	6	7	3		4	5			
	CONTROL UNIT																										
	ENTRADAS				Nemónico	SALIDAS																					
	OPCODE[3:0]	FUNC[23:20]	FUNC[19]	FUNC[15:14]	-	ALU_CONTROL[4:0]	WE_REG	SIZE	IMM_SRC[2]	IMM_SRC[1:0]	WE_MEM	BEQ	BNE	BLT	BGE	JUMP	W_REGV	W_MEMV	RESULT_SRC[1:0]	ALU_SRC[2:0]	NEATHER_PORTAL	NEATHER_RESET	NEATHER_MODE	NEATHER_WREG_SRC	TEA_SRC = ALU_CONTROL[0]	PC_ENABLE= (-) STALL	
0		0000	0	-	add		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0000	1	-	sub		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0001	0	-	sll		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0001	1	-	slt		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	IMM_SRC[2] = WE_REG OR WE_RVault
		0010	0	-	xor		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0010	1	-	srl		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
	0000	0011	0	-	sra	func[23:20],func[19]	1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0011	1	-	or		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0100	0	-	and		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0100	1	-	mul		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0101	0	-	div		1	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0101	1	-	SLEEP		0	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	1	
		0110	0	-	FREEZE		0	x	x	-	0	0	0	0	0	0	0	0	00	00	0	0	x	x	x	0	
1		-	-	00	addi	00000	1	x	1	10	0	0	0	0	0	0	0	0	00	11	0	0	x	x	x	1	
	0001	-	-	01	addiHIGH	00000	1	x	0	11	0	0	0	0	0	0	0	0	00	11	0	0	x	x	x	1	
		-	-	10	addiSigned	00000	1	x	1	11	0	0	0	0	0	0	0	0	00	11	0	0	x	x	x	1	
		-	-	11																		0					
2	0010	-	0	-	sw	00000	0	1	0	01	1	0	0	0	0	0	0	0	xx	11	0	0	x	x	x	1	
	0010	-	1	-	lw	00000	1	1	1	00	0	0	0	0	0	0	0	0	01	11	0	0	x	x	x	1	
3	0011	-	0	-	sb	00000	0	0	0	01	1	0	0	0	0	0	0	0	xx	11	0	0	x	x	x	1	
	0011	-	1	-	lb	00000	1	0	1	00	0	0	0	0	0	0	0	0	01	11	0	0	x	x	x	1	
4	0100	-	0	-	jal	xx	1	x	0	00	0	0	0	0	0	1	0	0	10 (ra = PC+4)	11	0	0	x	x	x	1	
4	0100	-	1	-	jalr	xx	0	x	1	01	0	0	0	0	0	1	0	0	00 (ra=ra+0)	11	0	0	x	x	x	1	
6		0000	0	-	beq	00001	0	x	0	10	0	1	0	0	0	0	0	0	xx	00	0	0	x	x	x	1	Z
		0000	1	-	bne	00001	0	x	0	10	0	0	1	0	0	0	0	0	xx	00	0	0	x	x	x	1	No Z
	0110	0001	0	-	blt	00001	0	x	0	10	0	0	0	1	0	0	0	0	xx	00	0	0	x	x	x	1	N
		0001	1	-	bge	00001	0	x	0	10	0	0	0	0	1	0	0	0	xx	00	0	0	x	x	x	1	N o Z
		0010	0	-	logvb	00001	0	x	0	10	0	0	1	0	0	0	0	0	xx	00	1	0	0	x	x	1	NEATHER_PORTAL resetea los regV además de permitir escritura en vault
5	0101																										
7	0111																										
8	1000																										
9	1001				NO USADA																						
10																											
13																											
11																											
12	1010	0001	0	-	sllv	00010	1	x	x	-	0	0	0	0	0	0	0	0	11 (TEA)	01	0	0	1	x	0	1	
		0010	1	-	slrv	00101	1	x	x	-	0	0	0	0	0	0	0	0	11 (TEA)	01	0	0	1	x	1	1	
	1011	0000	0	-	changev	00000	0	x	x	-	0	0	0	0	0	0	1	0	xx	00	0	0	1	0	x	1	
	1100	0000	0	-	closev	00000	0	x	x	-	0	0	0	0	0	0	0	0	xx	00	0	1	1	x	x	1	
	1110	-	0	-	swv	00000	0	x	0	01	0	0	0	0	0	0	0	1	xx	10	0	0	1	x	x	1	
14	1110	-	1	-	lwv	00000	0	x	1	00	0	0	0	0	0	0	1	0	xx	10	0	0	1	1	x	1	resultado de RAMV
		-	-	00	addiLOWv	00000	0	x	1	10	0	0	0	0	0	0	1	0	xx	11	0	0	1	0	x	1	
15	1111	-	-	01	addiHIGHv	00000	0	x	0	11	0	0	0	0	0	0	1	0	xx	11	0	0	1	0	x	1	